

Object Detection Models

YOLOv8, Faster R-CNN, DETR

Name	ID
Hour Mohamed	8119
Nadine Mohamed Tamish	8092
Seif Maged	8387

Contents

1	Architectures	3
1.1	YOLOV8	3
1.2	Faster R-CNN	3
1.3	DETR ([DE]tection [TR]ansformer)	5
2	Evaluation metrics	6
3	Results	6
3.1	YOLO	6
3.1.1	COCO Dataset	6
3.1.2	Pascal VOC Dataset	6
3.2	Faster R-CNN	7
3.2.1	COCO Dataset	7
3.2.2	Pascal VOC Dataset	7
3.3	DETR	7
3.3.1	COCO Dataset	7
3.3.2	Pascal VOC Dataset	8

1 Architectures

1.1 YOLOV8

- **Backbone:**
 - **CBS Blocks (Conv-BN-SiLU):** Basic building block: Convolution → BatchNorm → SiLU (Swish) activation.
 - **C2f Modules:**
 - * Similar to CSP (Cross Stage Partial) modules, but more efficient.
 - * Split input, process one part with bottlenecks, then fuse back.
 - **SPPF (Spatial Pyramid Pooling – Fast):**
 - * Helps to capture multi-scale features with minimal computation.
 - * Replaces traditional SPP with a faster variant.
 - **Downsampling via Strided Convolutions or MaxPool:**
 - * Reduces spatial resolution while increasing channel depth.
- **Neck (feature extractor):** Merges feature maps from different stages of the backbone to capture information at various scales.
- **Head:** Responsible for making predictions.

1.2 Faster R-CNN

- **Backbone (Feature Extractor):**
 - Typically a pre-trained CNN like ResNet-50 or ResNet-101.
 - Extracts convolutional feature maps from the input image.
 - Often includes Feature Pyramid Networks (FPN) to capture multi-scale features.
- **Region Proposal Network (RPN):**
 - A shallow network that slides over the feature map produced by the backbone.
 - For each location, it predicts:
 - * **Objectness score:** whether the region contains an object.
 - * **Bounding box offsets:** refinements to default anchor boxes.

- Outputs a set of region proposals (ROIs) with high objectness scores.
- **Anchor Boxes:**
 - Predefined boxes of different scales and aspect ratios used by the RPN.
 - Anchor boxes are refined using regression offsets predicted by the RPN.
- **ROI Align (Region of Interest Alignment):**
 - Takes variable-sized region proposals and extracts fixed-size (e.g., 7×7) feature maps from the backbone feature map.
 - ROI Align improves upon ROI Pooling by preserving spatial alignment (no rounding).
- **Head (Two Fully Connected Layers):**
 - Processes the aligned ROIs with two fully connected (FC) layers.
 - Outputs two branches:
 - * **Classification Branch:**
 - Uses a **Softmax** function.
 - Outputs class probabilities (including background).
 - * **Bounding Box Regression Branch:**
 - Outputs four coordinates for each class: $[t_x, t_y, t_w, t_h]$.
 - These are used to refine the final bounding box locations.
- **Post-processing:**
 - Applies **Non-Maximum Suppression (NMS)** to remove redundant overlapping boxes.
 - Final output is a set of object detections with class labels, confidence scores, and bounding boxes.
- **Training Strategy:**
 - Multi-task loss = classification loss (Softmax Cross-Entropy) + bounding box regression loss (Smooth L1).
 - Trained end-to-end in a two-stage pipeline:
 1. Train RPN to generate high-quality proposals.
 2. Train classification and regression heads on the proposed regions.

1.3 DETR ([DE]tection [TR]ansformer)

- **Backbone:**

- **ResNet-50:** Used as a feature extractor. It outputs a grid of image features that are passed into the transformer.
- **Positional Encoding:** Since transformers are permutation-invariant, positional encodings (sine and cosine based) are added to the backbone features to retain spatial information.

- **Transformer Encoder-Decoder:**

- **Encoder:** A standard transformer encoder processes the flattened feature map. It consists of multi-head self-attention layers and feed-forward networks.
- **Decoder:** Uses a fixed number of learned object queries. Each query attends to the encoder output and attempts to predict an object.
- **Object Queries:** Learnable embeddings that allow the model to detect a fixed number of objects in a single forward pass.

- **Prediction Head:**

- Consists of a shared feed-forward network applied to each decoder output.
- Each query outputs a bounding box and class label.
- DETR performs a bipartite matching (Hungarian algorithm) between predicted boxes and ground truth, enabling end-to-end optimization without anchor boxes or NMS.

- **Inference and Evaluation:**

- Pretrained DETR (ResNet-50) from Hugging Face was used.
- Inference was performed on the Pascal VOC 2007 test set.
- Predictions were post-processed using the built-in DETR image processor with a confidence threshold of 0.7.
- Results were converted to COCO format and evaluated using `pycocotools` for standard object detection metrics (AP, AR).

2 Evaluation metrics

- **IoU:** measures the overlap between predicted and ground truth bounding boxes
- **mAP:** measures how well the model detects and classifies objects.

3 Results

3.1 YOLO

3.1.1 COCO Dataset

Table 1: YOLO Performance on COCO Dataset

IoU Threshold	mAP	Precision	Recall
0.5:0.95	37.2	37.4	32.0
0.5	52.1	52.6	81.1

3.1.2 Pascal VOC Dataset

Table 2: YOLO Performance on Pascal VOC Dataset

Metric	Value
mAP@0.5	8.7
mAP@0.5:0.95	6.7
Precision	8.9
Recall	7.5

3.2 Faster R-CNN

3.2.1 COCO Dataset

Table 3: Faster R-CNN Performance on COCO Dataset

IoU Threshold	mAP	Precision	Recall
0.5:0.95	34.44	37.8	42.23
0.5	53.83	46.19	52.13

3.2.2 Pascal VOC Dataset

Table 4: Faster R-CNN Performance on Pascal VOC Dataset

Metric	Value
mAP@0.5	73.35
mAP@0.5:0.95	46.387
avg_IOU	0.511
Precision	17.77
Recall	93.52

3.3 DETR

3.3.1 COCO Dataset

Table 5: DETR Performance on COCO Dataset

IoU Threshold	mAP	Precision	Recall
0.5:0.95	36.5	39.0	42.1
0.5	52.3	46.5	49.7

3.3.2 Pascal VOC Dataset

Table 6: DETR Performance on Pascal VOC Dataset

Metric	Value
mAP@0.5	55.9
mAP@0.5:0.95	41.3
avg_IOU	62.6
Precision	55.9
Recall	47.0